

- **OpenVPN:** OpenVPN is a full-fledged, open source VPN solution for enterprises as well as individuals. Using OpenVPN's technology, we can deploy the certification, encryption, and authentication features of the OpenSSL library for protection. The VPN provides support for dynamic IP addresses and DHCP. OpenVPN is available for Android, macOS, Linux, Windows, and iOS.
- **Libreswan VPN:** Libreswan offers free and open source VPN software based on the popular IPsec and IKE standards of VPN protocols. Libreswan has been active for over 15 years.
- **SoftEther VPN:** SoftEther stands for 'Software Ethernet', and it is a free, open source VPN supporting multiple protocols. SoftEther VPN uses an ultra-optimised SSL-VPN protocol that delivers firewall resistance, low latency, and fast throughput. It can easily penetrate a network firewall with built-in NAT traversal. It supports SSL-VPN, L2TP, IPsec, EtherIP, OpenVPN, and L2TPv3 protocols.
- **Openswan VPN:** Openswan VPN is one of the most popular VPNs and is specially designed to cater to the network security demands of Linux users. It is an IPsec implementation for Linux and comes pre-installed in several Linux versions from Fedora, Ubuntu, Debian, Gentoo, and Red Hat.
- **Freelan VPN:** Freelan is a free, open source, multi-platform VPN software that abstracts a LAN over the internet. It works on Windows, Linux and Mac OS X.

Desktop virtualisation

Desktop virtualisation is a method of simulating a user workstation so that it can be accessed from a remotely connected device. By abstracting the user desktop in this way, organisations

can allow users to work from virtually anywhere with a network connection, using any desktop, laptop, tablet, or smartphone to access enterprise resources irrespective of the device or operating system employed by the remote user.

Remote desktop virtualisation is also a key component of digital workspaces. Virtual desktop workloads run on desktop virtualisation servers, which typically execute on virtual machines (VMs) either at on-premise data centers or in the public cloud. Since the user device is basically a display, keyboard, and mouse, a lost or stolen device presents a reduced risk to the organisation. All user data and programs exist in the desktop virtualisation server, and not on client devices. Some open source platforms/tools that enable desktop virtualisation are listed below.

Hypervisor: Hypervisor acts as a governing mechanism. It is a software layer that allows you to create and work on many virtual machines on a single hardware unit. There are several open source hypervisors available. Given below are a few popular ones.

- **Xen:** Xen is among the market leaders in the open source hypervisor arena. It supports the virtualisation of x86 IA64, and ARM, among other architectures. Xen has been used in the virtualisation of a wide array of guest OSs. A few operating systems supported by Xen are Windows, Solaris, and other versions of the BSD OS.
- **Linux KVM:** A kernel-based virtual machine (KVM) is a project based on GNU/Linux that has been developed for x86 computers. KVM is a full virtualisation solution for Linux on hardware containing virtualisation extensions. It provides the hardware virtualisation for a wide variety of guest operating systems, including Linux, Windows, macOS, ReactOS, and Haiku. Using

KVM, you can run multiple VMs on unmodified Linux or Windows images. Each VM has private virtualised hardware -- a network card, disk, graphics adapter, etc.

HCI: Hyper converged infrastructure (HCI) is fully software-defined IT infrastructure that virtualises all the elements of conventional 'hardware-defined' systems. It is essential for deploying modern virtual desktop infrastructure (VDI). A hyper converged environment is one where compute, storage, and network resources are all on the same servers, managed by a single interface.

We can run a full-featured virtualised infrastructure with optimal use of servers by using open source oVirt and Gluster to deploy a hyper converged solution. These two open source solutions work together to provide a hyper converged solution. Each server provides both hypervisor and storage capabilities, with the VM disk images stored on the Gluster volume. The Gluster volume allows storage from multiple servers to be presented as a single, unified name space to the client. It is configured from storage on the three servers as a replica-3 volume, which means every bit of data that's written is replicated to two other servers to provide redundancy and resiliency against server outages.

Red Hat Hyperconverged Infrastructure—based on leading OpenStack and virtualisation platforms—provides co-located, scalable, software-defined compute and storage, automated by the Red Hat Ansible Automation Platform on economical, industry-standard hardware.

RHV: Red Hat Virtualization (RHV) -- the virtualisation management solution to support server or desktop applications -- is an open, software-defined platform that virtualises Linux and